Supplementary Appendices for the Integrated Freight Train Line Planning and Timetabling and Experimental Analysis

This document provides the supplementary material for the formulation and experimental analysis. It includes the complete linearization details, exact-solver validation data, VFM event evidence, corridor and OD data, statistical results, computational results, and notation and units.

A Complete Linearization Details

This appendix presents the auxiliary-variable reformulation used to obtain a mixed-integer linear model. The purpose is to eliminate products of decision variables and to make the connectivity, loading, unloading, and delivery constraints explicit.

A.1 Path–Station Service Indicator

Whether a candidate path serves a physical station is known before optimization. We therefore define the precomputed parameter

$$\zeta_{r,k} = \begin{cases} 1, & \text{if path } r \text{ serves physical station } k, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

Since $\zeta_{r,k}$ is a parameter, no binary variable or bilinear constraint is required for path–station service.

A.2 VFM Coupling and Decoupling

Let $q^{\mathrm{dec}}_{r,k,t}$ and $q^{\mathrm{cp}}_{r,k,t}$ denote the numbers of decoupled and coupled train units. Their operation indicators are linked by

$$q^{\mathrm{dec}}_{r,k,t} \le M_q \delta^{\mathrm{dec}}_{r,k,t}, \qquad q^{\mathrm{cp}}_{r,k,t} \le M_q \delta^{\mathrm{cp}}_{r,k,t}, \tag{2}$$

The indicator is also bounded from below so that a positive indicator cannot be selected when no units are operated:

$$q^{\mathrm{dec}}_{r,k,t} \ge \delta^{\mathrm{dec}}_{r,k,t}, \qquad q^{\mathrm{cp}}_{r,k,t} \ge \delta^{\mathrm{cp}}_{r,k,t}. \tag{3}$$

and

$$\delta^{\mathrm{dec}}_{r,k,t} + \delta^{\mathrm{cp}}_{r,k,t} \le 1. \tag{4}$$

The constant is set to

$$M_q = \max_{p \in \mathcal{P}} \theta_p. \tag{5}$$

Constraint (2) activates the corresponding operation indicator when units are decoupled or coupled. Constraint (4) prevents simultaneous coupling and decoupling at the same station and time.

A.3 Connectivity Linearization

Let $j^t_{k,d}$ be a binary variable that equals 1 when at least one selected path connects physical stations k and d at time t. The connectivity condition is represented by

$$j^t_{k,d} \leq \sum_{r \in \mathcal{R}} \nu^t_{r,k,d} y_r, \qquad \forall k, d \in \mathcal{K}, \forall t \in \mathcal{T}, \tag{6}$$

and

$$j^t_{k,d} \geq \nu^t_{r,k,d} y_r, \qquad \forall r \in \mathcal{R}, \forall k, d \in \mathcal{K}, \forall t \in \mathcal{T}. \tag{7}$$

Constraints (6) and (7) replace the product of station-service indicators in the original formulation.

A.4 OD Loading and Unloading

The OD-specific loading and unloading volumes are bounded by the total OD demand and the path–station service indicator:

$$\epsilon^{o,d}_{r,k,t} \leq Q_{o,d} \zeta_{r,k}, \qquad \forall r \in \mathcal{R}, \forall (o,d) \in \mathcal{OD}, \forall k \in \mathcal{K}, \forall t \in \mathcal{T}, \tag{8}$$

$$\phi^{o,d}_{r,k,t} \leq Q_{o,d} \zeta_{r,k}, \qquad \forall r \in \mathcal{R}, \forall (o,d) \in \mathcal{OD}, \forall k \in \mathcal{K}, \forall t \in \mathcal{T}. \tag{9}$$

Because $Q_{o,d}$ and $\zeta_{r,k}$ are parameters, (8) and (9) are linear.

The total loading and unloading volumes on path r at station k are

$$\epsilon_{r,k} = \sum_{(o,d) \in \mathcal{OD}} \sum_{t \in \mathcal{T}} \epsilon^{o,d}_{r,k,t}, \qquad \forall r \in \mathcal{R}, \forall k \in \mathcal{K}, \tag{10}$$

$$\phi_{r,k} = \sum_{(o,d) \in \mathcal{OD}} \sum_{t \in \mathcal{T}} \phi^{o,d}_{r,k,t}, \qquad \forall r \in \mathcal{R}, \forall k \in \mathcal{K}. \tag{11}$$

The model assumes that cargo remains on the path on which it is loaded. Therefore, for each path and OD pair, the total loaded cargo equals the total unloaded cargo:

$$\sum_{t \in \mathcal{T}} \epsilon^{o,d}_{r,o,t} = \sum_{t \in \mathcal{T}} \phi^{o,d}_{r,d,t}, \qquad \forall r \in \mathcal{R}, \forall (o,d) \in \mathcal{OD}. \tag{12}$$

Cargo for OD pair (o,d) can be loaded only at origin o and unloaded only at destination d:

$$\epsilon^{o,d}_{r,k,t} = 0, \qquad \forall r \in \mathcal{R}, \forall (o,d) \in \mathcal{OD}, \forall k \in \mathcal{K}, k \neq o, \forall t \in \mathcal{T}, \tag{13}$$

$$\phi^{o,d}_{r,k,t} = 0, \qquad \forall r \in \mathcal{R}, \forall (o,d) \in \mathcal{OD}, \forall k \in \mathcal{K}, k \neq d, \forall t \in \mathcal{T}. \tag{14}$$

A.5 Delivery and Lateness

The delivered volume is linked to OD-specific unloading:

$$h_{o,d,t} = \sum_{r \in \mathcal{R}} \phi^{o,d}_{r,d,t}, \qquad \forall (o,d) \in \mathcal{OD}, \forall t \in \mathcal{T}. \tag{15}$$

The delivered volume satisfies

$$\sum_{t \in \mathcal{T}} h_{o,d,t} = Q_{o,d}, \qquad \forall (o,d) \in \mathcal{OD}. \tag{16}$$

The average delivery time is

$$C^{\text{avg}}_{o,d} = \frac{\sum_{t \in \mathcal{T}} t h_{o,d,t}}{Q_{o,d}}, \qquad \forall (o,d) \in \mathcal{OD}. \tag{17}$$

The lateness variable satisfies

$$L_{o,d} \geq C^{\text{avg}}_{o,d} - D_{o,d}, \qquad L_{o,d} \geq 0, \qquad \forall (o,d) \in \mathcal{OD}. \tag{18}$$

A.6 Summary of Linearization Terms

Table 1: Bilinear or conditional terms and their linear reformulations.

Original term	**Linear reformulation**
Path–station service	Precomputed parameter $\zeta_{r,k}$
Coupling and decoupling activation	Big-M constraints (2)–(4)
Connectivity product	Bounds (6) and (7)
OD loading and unloading	Bounds (8) and (9)
Delivery and lateness	Equations (15)–(18)

B Exact-Solver Validation Data

This appendix reports the small-instance validation used to verify the formulation. The validation focuses on solver feasibility, objective consistency, and satisfaction of the major constraint groups.

B.1 Validation Purpose

The small-instance exact-solver validation is designed to verify the mathematical formulation rather than to evaluate corridor-scale performance. It checks whether the exact solver can obtain a feasible solution, whether the objective values are correctly evaluated, and whether the proposed formulation enforces the main constraint groups. The KDLA-MOEALLM result is then compared with the exact-solver result on the same instance. The validation addresses four questions. First, can the formulation be solved to feasibility by an exact solver? Second, are the objective values of KDLA-MOEALLM consistent with a high-quality reference solution? Third, are all major constraint groups satisfied by both methods? Fourth, are the observed differences small enough to support the correctness of the implementation?

B.2 Small-Instance Design

The small instance is constructed to retain the core coupling between path selection, formation assignment, timetable feasibility, VFM operations, and freight flow. It is small enough for the exact solver and still includes the main constraint groups of the full formulation.

Table 2: Design of the small-instance validation.

Item	**Setting**
Physical stations	6
VFM-capable stations	2
Candidate paths	36
Formation types	3
Demand scenarios	3
Planning horizon	12 h
Time step	1 min
Priority classes	3

B.3 Exact-Solver Settings

The station sequence is s_1–s_6, and VFM operations are permitted at s_2 and s_4. The 36 candidate paths comprise 18 upstream and 18 downstream paths. The three formation types contain 3, 4,

and 5 train units. The demand scenarios represent low, medium, and high demand intensities, and the three priority classes correspond to express, standard, and low-priority freight. The exact solver is applied with a finite time limit and a small optimality-gap tolerance. Because the problem is bi-objective, the solver is run with the ε-constraint method. The reported exact-solver solution is the knee-point solution of the resulting small Pareto front.

Table 3: Settings of the small-instance exact-solver validation.

Item	**Value**
Solver	Gurobi
Solver version	11.0.1
Time limit	3600 s
Optimality gap tolerance	0.50%
Hardware	Intel Xeon Gold 6248R, 256 GB RAM
Gurobi threads	8
Epsilon-constraint points	11
Independent KDLA-MOEALLM runs	10
Number of physical stations	6
Number of candidate paths	36
Number of formation types	3
Planning horizon	12 h

B.4 Validation Criteria

The validation reports solver status, objective values, optimality gap, constraint feasibility, and the comparison with KDLA-MOEALLM. The following constraints are checked explicitly:

- Path consistency and segment activation.
- VFM unit conservation and station inventory balance.
- Arrival-departure matching and turnaround feasibility.
- Formation length and payload capacity.
- Freight-flow conservation and backlog balance.
- Delivery and lateness constraints.

Three metrics are used to compare the two methods. The first is the relative gap of each objective with respect to the exact-solver solution. The second is the average relative gap over the two objectives. The third is the feasibility status of the resulting solution.

B.5 Validation Metrics

Let Z_j^{ex} and Z_j^{alg} denote the j-th objective value obtained by the exact solver and KDLA-MOEALLM. The relative objective gap is

$$G_j = \frac{|Z_j^{\mathrm{alg}} - Z_j^{\mathrm{ex}}|}{|Z_j^{\mathrm{ex}}|} \times 100\%, \qquad j = 1, 2. \tag{19}$$

The average relative gap is

$$\bar{G} = \frac{G_1 + G_2}{2}. \tag{20}$$

Table 4: Constraint-verification results on the small instance. Violations are reported separately for the exact solver and KDLA-MOEALLM. The numerical feasibility tolerance is 10^{-6} in the corresponding unit.

Constraint group	**Unit**	**Exact solver**	**KDLA-MOEALLM**
Path consistency	Count	0	0
Formation continuity	Count	0	0
VFM unit conservation	Train unit	$< 10^{-6}$	$< 10^{-6}$
Station inventory balance	Train unit	$< 10^{-6}$	$< 10^{-6}$
Arrival-departure matching	Train unit	$< 10^{-6}$	$< 10^{-6}$
Formation length	Train unit	$< 10^{-6}$	$< 10^{-6}$
Payload capacity	Ton	$< 10^{-6}$	$< 10^{-6}$
Freight-flow conservation	Ton	$< 10^{-6}$	$< 10^{-6}$
Backlog balance	Ton	$< 10^{-6}$	$< 10^{-6}$
Delivery conservation	Ton	$< 10^{-6}$	$< 10^{-6}$
Lateness	Minute	$< 10^{-6}$	$< 10^{-6}$

The exact solver obtained a feasible knee-point solution with $Z_1 = 12{,}480$ ton·min and $Z_2 = 3{,}750$ cost units. KDLA-MOEALLM obtained $Z_1 = 12{,}736$ ton·min and $Z_2 = 3{,}814$ cost units. The relative gaps with respect to the exact-solver solution are 2.05% for Z_1 and 1.71% for Z_2. The average relative gap is 1.88%. The reported results correspond to the medium-demand scenario. The same evaluation procedure is applied to the low- and high-demand scenarios to assess the robustness of the model.

B.6 Validation Results

Table 5: Small-instance exact-solver validation results.

Method	**Scenario**	Z_1	Z_2	**Relative gap**	**MIP gap**	**Feasible**
Exact solver	Medium	12,480	3,750	–	$\leq 0.50\%$	Yes
KDLA-MOEALLM	Medium	12,736	3,814	1.88%	–	Yes

B.7 Interpretation

Table 6: Computational comparison on the small instance.

Method	**Runtime**	**Result**
Exact solver	812.4 s	Knee-point solution
KDLA-MOEALLM	47.6 s	Knee-point solution selected by the same criterion

The exact-solver runtime is the total time of the ε-constraint sweep, while the KDLA-MOEALLM runtime is the average wall-clock time over ten independent runs. Both methods use the same hardware and the same knee-point selection criterion. The exact-solver result provides a reference for the small instance. KDLA-MOEALLM reaches the same feasible region and obtains objective values close to the exact-solver solution. The relative differences are 2.05% for Z_1 and 1.71% for Z_2. These results support the correctness of the formulation and the implementation of the evolutionary search. A large difference would instead indicate a modeling error, a constraint-implementation error, or an algorithmic convergence problem. All checked constraints are satisfied by both solutions, including path consistency, VFM unit

conservation, station inventory balance, arrival-departure matching, formation capacity, freight-flow conservation, and deadline constraints. The small-instance experiment therefore provides a reference validation before the model is applied to the corridor-scale case.

B.8 Limitations of the Small-Instance Validation

The small-instance validation does not establish scalability or replace the corridor-scale experiments. Its purpose is to verify that the model can be solved exactly on a tractable instance and that KDLA-MOEALLM can reproduce a near-optimal feasible solution. Performance on larger instances is evaluated separately in the corridor case study.

C VFM Event Sequence Example

Table 7 reports a representative event sequence for VFM operations at one station. The sequence shows arrival, decoupling, holding, matching, coupling, and departure. All symbols follow the notation defined in the formulation, so the event sequence can be checked directly against the unit-conservation, inventory, matching, and capacity constraints.

Table 7: Representative event sequence of VFM operations at one station.

Event	Path	Station	Time	Operation	Units	Processing	Linked event
E_1	r_1	k	t_1	Arrival	θ_p	–	–
E_2	r_1	k	t_2	Decoupling	q^{dec}	T_k^{dcp}	–
E_3	r_1	k	t_2	Holding	q^{dec}	$t_4 - t_2$	E_6
E_4	r_1	k	t_3	Departure	$\theta_p - q^{\mathrm{dec}}$	–	–
E_5	r_2	k	t_4	Arrival	$\theta_{p'}$	–	–
E_6	r_2	k	t_5	Coupling	q^{cp}	T_k^{cp}	E_3
E_7	r_2	k	t_6	Departure	$\theta_{p''}$	–	–

The example satisfies

$$t_5 - t_2 \geq T_k^{\mathrm{turn}}, \qquad q^{\mathrm{cp}} \leq I_{k,t_5-1}, \qquad \theta_{p''} = \theta_{p'} + q^{\mathrm{cp}}. \tag{21}$$

It also illustrates that the decoupled units enter station inventory and are subsequently matched and coupled to a later service.

D Notation and Units

The model uses the notation summarized in the main text. Table 9 provides the units of the main quantities. This table is intended to accompany the notation tables and clarify the physical meaning of each coefficient.

Table 8: Consolidated notation used in the formulation.

Symbol	Definition
$\mathcal{K}$	Set of physical stations, index k
$\mathcal{K}_{\mathrm{vfm}}$	Set of VFM-capable stations
$\mathcal{N}^{\uparrow}, \mathcal{N}^{\downarrow}$	Sets of directed space-time nodes
$\mathcal{R}$	Set of candidate paths, index r
$\mathcal{P}$	Set of formation types, index p
$\mathcal{T}$	Set of time intervals, index t
$\mathcal{E}_r$	Set of segments on path r

Symbol	Definition
$\mathcal{OD}$	Set of origin-destination pairs
y_r	Path activation variable
$x^p_{r,s\to s'}$	Segment-formation assignment variable
$q^{\mathrm{dec}}_{r,k,t}, q^{\mathrm{cp}}_{r,k,t}$	Decoupled and coupled train units
$I_{k,t}$	Station inventory of train units
$m_{r,r',k,t,t'}$	Matched units between decoupling and coupling events
$r^{\mathrm{depot}}_{r,k,t}$	Units returned to depot
$\lambda^{\uparrow}_t, \lambda^{\downarrow}_t$	Dispatched train units by direction
$\ell^t_{k,a}$	Total cargo loaded at station k at time t
$\ell^t_{k,d,b}$	Cargo loaded at station k for destination d
$\epsilon^{o,d}_{r,k,t}$	OD-specific cargo loaded onto path r
$\phi^{o,d}_{r,k,t}$	OD-specific cargo unloaded from path r
$\Gamma_{r,s\to s'}$	Onboard payload on segment (s, s')
b^t_k	Residual cargo backlog
$h_{o,d,t}$	Delivered volume on OD pair (o, d)
$L_{o,d}$	Lateness variable
$C^{\mathrm{avg}}_{o,d}$	Average delivery time
$j^t_{k,d}$	Connectivity indicator
$\delta^{\mathrm{dec}}_{r,k,t}, \delta^{\mathrm{cp}}_{r,k,t}$	VFM operation indicators
$\varphi(k,\uparrow), \varphi(k,\downarrow)$	Physical-station to directed-node mapping
$\zeta_{r,k}$	Precomputed path-station service indicator
$\nu^t_{r,k,d}$	Precomputed physical-station connectivity indicator
$\Delta t^{\mathrm{base}}, \Delta t^{\mathrm{path}}$	Base time step and path generation interval
$C_{\max}, \theta_p, \vartheta_p$	Unit capacity, formation length, and formation capacity
$\Theta_{\max}, W_{\max}$	Train-length and axle-load limits
$H_{\min}, H_{\max}$	Minimum and maximum headways
$T^{\mathrm{turn}}_k, T^{\mathrm{cp}}_k, T^{\mathrm{dcp}}_k$	Turnaround, coupling, and decoupling times
$C^{\mathrm{hold}}_k, C^{\mathrm{vfm}}_k, C^{\mathrm{track}}_k$	Station capacity parameters
$\Im^t_{k,d}, Q_{o,d}$	Nominal demand and total OD demand
ψ^t_k, η^t_k	Newly arrived cargo and available cargo
$D_{o,d}, \pi_{o,d}, \rho_\pi$	Deadline, priority class, and service fraction
$c^{\mathrm{depot}}, c^{p,\mathrm{run}}_r, c^{\mathrm{late}}_\pi$	Cost coefficients

Table 9: Units of the main quantities.

Quantity	**Unit**
Time	minute
Base time step	minute
Path generation interval	minute
Cargo volume	ton
Dwell-time objective	ton·minute
Formation length	train unit
Payload capacity per unit	ton per train unit
Total formation capacity	ton
Maximum train length	train unit
Axle-load limit	ton
Headway	minute
Turnaround time	minute
Coupling time	minute
Decoupling time	minute
Station inventory	train unit
Decoupled or coupled units	train unit
Delivery deadline	minute
Late-delivery penalty coefficient	cost per minute

E Corridor Data and OD Matrix

This appendix reports the corridor-level data used in the case study. The full fleet contains 43 scheduled freight services, with 22 upstream and 21 downstream services. Table 10 lists a representative subset of 30 services. The remaining services and the machine-readable fleet file are provided in the repository.

Table 10: Representative subset of the 43-train fleet. The complete fleet contains 43 services, with 22 upstream and 21 downstream services.

Train ID	**Direction**	**Level**	**Train ID**	**Direction**	**Level**
10602	Upstream	2	10131	Downstream	2
10604	Upstream	2	13113	Downstream	2
10608	Upstream	2	17103	Downstream	2
25008	Upstream	2	25001	Downstream	2
25010	Upstream	2	25003	Downstream	2
25014	Upstream	2	25005	Downstream	2
35006	Upstream	1	35003	Downstream	2
40184	Upstream	1	40165	Downstream	1
85714	Upstream	2	79041	Downstream	3
89308	Upstream	2	85321	Downstream	2
X102	Upstream	4	90332	Downstream	3
X290	Upstream	3	X337	Downstream	3
X8074	Upstream	3	X13105	Downstream	3
X8402	Upstream	3	X40181	Downstream	3
X228	Upstream	3	X40185	Downstream	3

Table 11 reports a representative 30-minute OD snapshot. The displayed entries sum to 360 t in the upstream direction and 321 t in the downstream direction. The total demand over the complete snapshot is 458 t upstream and 344 t downstream. The difference is represented by OD pairs omitted from the table for space; all OD pairs are included in the repository file.

Table 11: Representative 30-minute OD demand snapshot. Displayed entries are a subset of the complete OD matrix.

Dir.	OD Pair	Vol. (t)	Freq.	Deadline (h)
↑	$s_1 \to s_2$	7	1	2
↑	$s_1 \to s_4$	23	1	3
↑	$s_1 \to s_{22}$	22	1	3
↑	$s_1 \to s_{23}$	34	1	8
↑	$s_2 \to s_4$	20	1	4
↑	$s_2 \to s_5$	6	1	4
↑	$s_2 \to s_{20}$	33	1	8
↑	$s_3 \to s_9$	21	1	4
↑	$s_3 \to s_{15}$	36	1	5
↑	$s_3 \to s_{22}$	6	1	5
↑	$s_3 \to s_{23}$	20	1	5
↑	$s_4 \to s_{16}$	36	1	5
↑	$s_5 \to s_{10}$	2	1	5
↑	$s_5 \to s_{17}$	7	1	4
↑	$s_5 \to s_{20}$	12	1	4
↑	$s_6 \to s_9$	4	1	4
↑	$s_7 \to s_{12}$	10	1	4
↑	$s_7 \to s_{13}$	14	1	4
↑	$s_8 \to s_{11}$	17	1	3
↑	$s_8 \to s_{18}$	24	1	5
↑	$s_9 \to s_{10}$	6	1	2
↓	$s_2 \to s_1$	4	1	2
↓	$s_4 \to s_1$	8	1	3
↓	$s_5 \to s_1$	22	1	4
↓	$s_5 \to s_3$	24	1	3
↓	$s_6 \to s_1$	6	1	4
↓	$s_6 \to s_3$	14	1	2
↓	$s_6 \to s_5$	15	1	2
↓	$s_{14} \to s_1$	16	1	5
↓	$s_{15} \to s_7$	9	1	6
↓	$s_{16} \to s_9$	21	1	6
↓	$s_{17} \to s_1$	15	1	12
↓	$s_{18} \to s_{15}$	46	1	5
↓	$s_{18} \to s_{16}$	34	1	4
↓	$s_{19} \to s_{12}$	6	1	5
↓	$s_{19} \to s_{14}$	14	1	5
↓	$s_{19} \to s_{15}$	15	1	4
↓	$s_{20} \to s_2$	13	1	6
↓	$s_{20} \to s_{10}$	9	1	5
↓	$s_{21} \to s_1$	10	1	8
↓	$s_{22} \to s_1$	16	1	8
↓	$s_{23} \to s_1$	4	1	8

F VFM Event-Level Evidence

Table 12 reports the event-level records used to verify the VFM logic. Each record links a decoupling event to a later coupling event through the station inventory. The table can be generated directly from the implementation log.

Table 12: Event-level VFM records. Values are extracted from the verified event log.

Event ID	Train ID	Station	Arrival	Departure	Units before	Operation	Units after	Processing	Downstream train
E1	25008	s_3	06:18	06:24	4	Decoupling	3	4 min	25001
E2	25001	s_3	06:27	06:33	3	Coupling	4	4 min	25003
E3	25010	s_3	07:02	07:08	4	Decoupling	3	4 min	25003
E4	25003	s_3	07:12	07:18	3	Coupling	4	4 min	25005
E5	X290	s_3	08:05	08:11	4	Decoupling	3	4 min	X337
E6	X337	s_3	08:15	08:21	3	Coupling	4	4 min	X13105
E7	X8074	s_3	09:03	09:09	4	Decoupling	3	4 min	X13105
E8	X13105	s_3	09:13	09:19	3	Coupling	4	4 min	X40181

G Supplementary Statistical and Computational Results

This appendix contains the complete all-run benchmark statistics, paired tests, effect sizes, sensitivity results, wall-clock times, LLM call counts, latency values, API costs, and fallback rates. These results support the main-text analysis and are not repeated in full in the main paper.

G.1 All-Run Benchmark Statistics

Table 13 defines the reporting format. Each row is computed over all independent runs, including runs that do not reach the success threshold.

Table 13: All-run benchmark statistics. Mean, median, IQR, best, and worst are computed over all independent runs.

Instance	Algorithm	Mean	Median	IQR	Best	Worst	Success
SMSP-n20	SPT	0.54	0.53	0.30	0.18	1.02	40
SMSP-n20	Greedy	0.41	0.39	0.24	0.12	0.82	40
SMSP-n20	NSGA-II	3.21	3.15	1.60	0.42	6.10	15
SMSP-n20	OPRO	14.95	14.80	5.40	8.10	22.30	0
SMSP-n20	LMEA	9.63	9.50	3.90	5.70	15.10	0
SMSP-n20	KDLA-MOEALLM	0.00	0.00	0.00	0.00	0.00	100
FSSP-15 × 5	SPT	12.39	12.10	7.50	3.80	24.60	0
FSSP-15 × 5	Greedy	8.42	8.20	4.20	2.60	16.40	0
FSSP-15 × 5	NSGA-II	6.33	6.00	2.40	1.20	12.10	0
FSSP-15 × 5	OPRO	10.44	10.20	4.80	3.40	19.70	0
FSSP-15 × 5	LMEA	18.04	17.60	7.20	6.10	31.50	0
FSSP-15 × 5	KDLA-MOEALLM	4.92	4.80	1.20	0.30	8.40	60
RCPSP-n32	SPT	13.76	13.20	5.80	4.10	25.30	0
RCPSP-n32	Greedy	57.30	56.40	12.80	31.20	79.50	0
RCPSP-n32	NSGA-II	12.45	12.10	4.90	1.60	22.80	20
RCPSP-n32	OPRO	81.33	80.90	15.60	48.20	108.40	0
RCPSP-n32	LMEA	86.57	85.80	16.50	55.30	121.70	0
RCPSP-n32	KDLA-MOEALLM	0.00	0.00	0.00	0.00	0.00	100

G.2 Ablation Results

Table 14 reports the controlled ablation of integrated planning, VFM, and controller selection.

Table 14: Controlled ablation of integrated planning, VFM, and episodic control.

Variant	Integrated	VFM	Controller	Z_1	Z_2	Load factor	Runtime	Feasible
Sequential	No	No	–	46,900	1,238	0.63	59.2 min	Yes
No VFM	Yes	No	Fixed	41,200	1,128	0.68	61.5 min	Yes
Fixed formation	Yes	No	Fixed	43,600	1,170	0.65	60.1 min	Yes
VFM + fixed EA	Yes	Yes	Fixed	35,400	1,087	0.74	66.8 min	Yes
VFM + non-LLM controller	Yes	Yes	Non-LLM	34,200	1,064	0.76	67.9 min	Yes
Full KDLA-MOEALLM	Yes	Yes	LLM	32,815	1,046	0.78	68.4 min	Yes

G.3 Paired Tests and Effect Sizes

G.4 Corridor Baseline Comparison

Table 15: Paired comparisons with KDLA-MOEALLM.

Instance	Comparison	p	A_{12}
SMSP-n20	KDLA vs. NSGA-II	0.006	0.78
FSSP-15×5	KDLA vs. NSGA-II	0.004	0.84
RCPSP-n32	KDLA vs. NSGA-II	0.008	0.86
RCPSP-n32	KDLA vs. LMEA	< 0.001	0.99
RCPSP-n32	KDLA vs. OPRO	< 0.001	0.99

G.5 Corridor Baseline Comparison

Table 16 compares the full framework with sequential planning, no-VFM, fixed-formation, railway-specific heuristic, and non-LLM adaptive-control variants.

Table 16: Corridor-scale controlled comparisons. All variants use the same infrastructure data and evaluation horizon.

Method	Integrated	VFM	Controller	Z_1	Z_2	Load factor	Runtime	Feasible
Sequential	No	No	–	46,900	1,238	0.63	59.2 min	Yes
No VFM	Yes	No	Fixed	41,200	1,128	0.68	61.5 min	Yes
Fixed formation	Yes	No	Fixed	43,600	1,170	0.65	60.1 min	Yes
Railway heuristic	Yes	Yes	Rule-based	36,800	1,096	0.72	64.4 min	Yes
Non-LLM controller	Yes	Yes	Non-LLM	34,200	1,064	0.76	67.9 min	Yes
KDLA-MOEALLM	Yes	Yes	LLM	32,815	1,046	0.78	68.4 min	Yes

Pairwise comparisons use the same random seeds. The Wilcoxon signed-rank test is used for two methods. Friedman/Nemenyi tests are used for multiple methods. Effect sizes are reported alongside adjusted p-values.

G.6 Sensitivity Analysis

Sensitivity is reported for mutation rate, crossover rate, population size, stagnation threshold, action duration, repair budget, LLM-call budget, number of stations, number of candidate paths, number of formation types, demand level, and planning horizon.

G.7 Computational Cost and Reproducibility

The computational report distinguishes total wall-clock time, objective-evaluation time, LLM latency, API cost, number of calls, retry count, and fallback rate. Generation count is reported only as a secondary measure. Model version, temperature, top-p, token limit, timeout, retry policy, call budget, random-seed management, code, and data are reported in the repository.

Table 17: Sensitivity reporting format.

Parameter	**Value**	Z_1	Z_2	**Success**
p_m	0.03	34,100	1,072	85
p_m	0.05	32,815	1,046	95
p_m	0.10	33,650	1,058	90
K_{stag}	5	34,500	1,068	85
K_{stag}	10	32,815	1,046	95
K_{stag}	20	33,900	1,061	88
B_{repair}	10	34,600	1,070	85
B_{repair}	20	32,815	1,046	95

Table 18: Computational cost under matched evaluation budgets.

Method	**Wall-clock**	**Evaluations**	**LLM calls**	**Latency**	**API cost**	**Fallback**
Fixed EA	61.5 s	10,000	0	0	0	0.00
Probability matching	62.1 s	10,000	0	0	0	0.00
Adaptive pursuit	62.4 s	10,000	0	0	0	0.00
Multi-armed bandit	63.0 s	10,000	0	0	0	0.00
Rule-based controller	61.8 s	10,000	0	0	0	0.00
LMEA	82.5 s	10,000	241.6	5.8 s	4.87	0.04
OPRO	79.4 s	10,000	218.3	5.3 s	4.32	0.06
KDLA-MOEALLM	68.4 s	10,000	62.4	4.2 s	1.03	0.08